Juan David Carvajal-Agudelo

PhD Student, Veterinary Medical Sciences · University of Calgary

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scholar.google.com/citations?user=_loX9b4AAAAJ

SUMMARY

Molecular biologist from Colombia and PhD student in Veterinary Medical Sciences (Wildlife Health and Ecology) at the University of Calgary. My doctoral project studies emerging pathogens in wild mammals from Costa Rica using molecular and bioinformatic approaches and artificial intelligence. My background spans molecular biology, bioinformatics, the microbiome, tick-borne pathogens and space biology, with experience in molecular diagnostics, field sampling of wild mammals, zebrafish research on bone under simulated microgravity, and university teaching.

Research areas: Microgravity Biology, Microbiome, Tick-borne Pathogens, Bioinformatics, Metagenomics, RT-qPCR, Biodiversity, STEM Education, Phylogenetics & Evolution, Disease Ecology, Mammalogy, Animal Welfare, Genomics, Transcriptomics, Molecular Diagnostics, Space Biology.

EDUCATION

Doctor of Philosophy in Veterinary Medical Sciences
University of Calgary, Alberta, Canada
Started 2026
Specialization in Wildlife Health and Ecology.
Project: Emerging pathogens in wild mammals from Costa Rica.

Master in Biological Sciences
Universidad de Caldas, Colombia
2022
Thesis: Bacterial microbiome of *Ornithodoros cf. hasei* (Acari: Argasidae) and wild mammals in Arauca, Colombian Orinoquia.

Exchange Program
Mount Allison University, Sackville, NB, Canada
2021
Project: Novel *Borrelia* (Spirochaetales) genotypes in *Ornithodoros hasei* (Schulze, 1935) (Acari: Argasidae) and wild mammals from the Orinoco River basin of Colombia.

Bachelor of Science, Biology
Universidad de Caldas, Colombia
2018
Research participation: Expression of HER2, MUC1 and ER-α genes in the early diagnosis of canine breast cancer.

PROFESSIONAL EXPERIENCE

Graduate Student (PhD), Wildlife Health and Ecology
Faculty of Veterinary Medicine, University of Calgary, Alberta, Canada
2026 – Present
Doctoral research on emerging pathogens in wild mammals from Costa Rica, supported by an NSERC Canada Graduate Scholarship.

Junior Researcher, Investigador Junior
Ministerio de Ciencia, Tecnología e Innovación (Minciencias), Colombia
May – July 2026
Officially recognized within the National System of Science, Technology and Innovation (SNCTel) of Colombia. Scientific evaluator for Convocatoria 975 de 2026, reviewing master's-level research proposals.

Laboratory Instructor, Genes and Genomes Lab
Mount Saint Vincent University, Halifax, NS, Canada
Fall 2025
Taught laboratory sessions for biology major students in molecular biology and microbiology techniques: cloning, PCR, DNA extraction, gel electrophoresis, and related methods. Designed and delivered hands-on instruction aligned with the undergraduate biology curriculum.

Research Assistant
Bone Development Lab, Mount Saint Vincent University, Halifax, NS, Canada
February 2022 – November 2025

- Managed zebrafish husbandry and maintained welfare in accordance with CCAC guidelines.
- Wrote and assisted with Animal Care Committee (ACC) protocols.
- Supported CSA-funded research on bone physiology via simulated microgravity and vibration experiments; techniques included gene expression analysis, tissue staining, cell culture, and advanced microscopy.
- Conducted RT-qPCR, statistical analysis, prepared reports, and contributed to scientific articles and grant applications.
- Supported bioinformatics work in transcriptomics and data analysis.
- Supervised and mentored undergraduate and graduate students.

Molecular Analyst, SARS-CoV-2 Detection
Dirección Territorial de Salud de Caldas & Universidad de Caldas, Colombia
Jul–Dec 2020 / Jun–Dec 2021

- Reception, handling and molecular processing of COVID-19 samples, including nasopharyngeal swabs, in a certified molecular biology laboratory (BSL-2).
- Viral RNA extraction and RT-qPCR detection of SARS-CoV-2.
- Assisted with molecular biology laboratory protocols.
- Applied biosafety and biohazard waste management procedures.

Research Assistant
Genetics Laboratory, Universidad de Caldas, Colombia
March 2019 – March 2021

- Conducted field work collecting samples from wild mammals for pathogen and microbiome studies.
- Contributed to disease ecology research on pathogens, vectors and their wild mammal hosts.
- Pathogen detection via PCR and sequencing with molecular markers.
- 16S metagenomic bioinformatic analysis; report and article preparation.
- Specimen preparation and taxidermy support at the Museum of Natural History (mammalogy).

Research Internship, Lloyd's Tick Lab
Mount Allison University, Sackville, NB, Canada
Feb – Jun 2021
Multilocus DNA methodology for *Borrelia* sp. genotype characterization. Funded by Global Affairs Canada, ELAP Scholarship.

Biology Tutor Expert
NextLearning / Chegg, Online
April – July 2020
Problem solving and teaching in biological sciences on a professional online tutoring platform.

Laboratory Monitor (Teaching Assistant)
Molecular Biology and Cell Biology Laboratories, Universidad de Caldas, Colombia
February 2017 – February 2018
Teaching assistant for both the Molecular Biology and Cell Biology laboratory courses. Instruction in electrophoresis, PCR, RT-qPCR, DNA/RNA extraction, histology, and microscopy. Evaluation and monitoring of undergraduate biology students.

PUBLICATIONS

2026 **Transcriptomic profiling and RANKL/RANK/OPG-mediated osteoclastogenesis in zebrafish larvae under simulated microgravity conditions**. **Carvajal-Agudelo, J. D.**, & Franz-Odendaal, T. A. *Frontiers in Cell and Developmental Biology, 14, 1786373.* doi:10.3389/fcell.2026.1786373

2026 **Differential effects of vibration on bone tissue in zebrafish**. **Carvajal-Agudelo, J. D.**, & Franz-Odendaal, T. A. *Microgravity Science and Technology, 38(4), 43.* doi:10.1007/s12217-026-10263-2

2026 **Disrupted osteogenic gene expression of osteoblasts in vivo under simulated microgravity: vibration exposure as a potential countermeasure**. LeDrew, E., **Carvajal-Agudelo, J.**, & Franz-Odendaal, T. A. *Life Sciences in Space Research.* doi:10.1016/j.lssr.2026.03.004

2026 **20 Years of the ISCB Student Council Symposium: shaping computational biology and future leaders**. Fernandez-Martin, M., Castillo-Orozco, A., Poley-Gil, M., Pose-Lagoa, I., Urquiza-Zurich, S., Hayoun-Mya, O., Machado-Reyes, D., Treccani, M., Veschetti, L., **Carvajal-Agudelo, J. D.**, Boll, L. M., Garcia-Valiente, A., Parra, R. G., & Olguín-Orellana, G. J. *Bioinformatics Advances, 6(1), vbag097.* doi:10.1093/bioadv/vbag097

2025 **In Vitro Culturing Technique for Studying Cellular Dynamics in Zebrafish Scales**. **Carvajal-Agudelo, J. D.** & Franz-Odendaal, T. A. *Journal of Visualized Experiments (JoVE), (215), e66619.* doi:10.3791/66619

2025 **BMP and WNT signaling are involved in tracheal cartilage development in chicken embryos**. Muntzar, R., **Carvajal-Agudelo, J. D.**, Tench, L., Apienti, T. A. O., & Franz-Odendaal, T. A. *Developmental Biology.* doi:10.1016/j.ydbio.2025.04.020

2024 **Teaching zebrafish development in a STEM-based science camp for middle school students**. Franz-Odendaal, T. A., **Carvajal-Agudelo, J. D.** & Marchand, S. *Zebrafish, 21(2), 92–100.* doi:10.1089/zeb.2023.0082

2024 **Reduced ossification caused by 3D simulated microgravity exposure is short-term in larval zebrafish**. **Carvajal-Agudelo, J. D.**, Eaton, J., & Franz-Odendaal, T. A. *Life Sciences in Space Research, 41, 127–135.* doi:10.1016/j.lssr.2024.02.006

2023 **Effects of simulated microgravity and vibration on osteoblast and osteoclast activity in cultured zebrafish scales**. **Carvajal-Agudelo, J. D.**, McNeil, A., & Franz-Odendaal, T. A. *Life Sciences in Space Research, 38, 39–45.* doi:10.1016/j.lssr.2023.05.002

2023 **Comparison of effects of 2D and 3D simulated microgravity rotation on ossification in larval Danio rerio**. Eaton, J., **Carvajal-Agudelo, J. D.**, & Franz-Odendaal, T. A. *Microgravity Science and Technology, 35(5), 53.* doi:10.1007/s12217-023-10077-6

2023 **Turn on the lights! Fundamental science leads to scientific progress: a perspective from developmental biology**. Franz-Odendaal, T. A., **Carvajal-Agudelo, J. D.**, Apienti, T. A. O., Drake, P. M., Eaton, J., McInnis, S., & Muntzar, R. *Proceedings of the Nova Scotian Institute of Science (NSIS), 53(1).* View article

2023 **Morphological and molecular confirmation of *Ornithodoros hasei* (Schulze, 1935) in Colombia**. Ossa-López, P. A., Mancilla-Agrono, L. Y., Banguero-Micolta, L. F., Ramírez-Chaves, H. E., **Carvajal-Agudelo, J. D.**, Muñoz-Leal, S., Labruna, M. B., Lloyd, V., & Rivera-Páez, F. A. *Ticks and Tick-borne Diseases.* doi:10.1016/j.ttbdis.2023.102142

2022 **Bacterial and tick-borne pathogen assemblages in *Ornithodoros cf. hasei* and wild mammal hosts, Orinoquia, Colombia**. **Carvajal-Agudelo, J. D.**, Ramírez-Chaves, H. E., Ossa-López, P. A. & Rivera-Páez, F. A. *Experimental and Applied Acarology.* doi:10.1007/s10493-022-00724-9

2022 **Evaluation of reference genes for qRT-PCR using blood samples in canine mammary research**. **Carvajal-Agudelo, J. D.**, Rivera-Páez, F. A., Perez-Zapata, J. M., Ossa-López, P. A., Morales-Álvarez, E. D., Giraldo-Chalarca, L., & Cortes-Mera, D. M. *Veterinarska Stanica, 53(1).* doi:10.46419/vs.53.1.5

2021 **Ticks (Acari: Ixodoidea) associated with mammals in Colombia: historical review, molecular confirmation, and new relationships**. Ortíz-Giraldo, M., Tobón-Escobar, W. D., Velásquez-Guarín, D., Usma-Marín, M. F., Ossa-López, P. A., Ramírez-Chaves, H. E., **Carvajal-Agudelo, J. D.**, & Rivera-Páez, F. A. *Parasitology Research, 1–12.* doi:10.1007/s00436-020-06989-6

2021 **Field blood preservation and DNA extraction from wild mammals: methods for biodiversity studies**. **Carvajal-Agudelo, J. D.**, Trujillo-Betancur, M. P., Velásquez-Guarín, D., Ramírez-Chaves, H. E., Pérez-Cárdenas, J. E., & Rivera-Páez, F. A. *Revista UDCA Actualidad & Divulgación Científica, 24(1).* doi:10.31910/rudca.v24.n1.2021.1766

2021 **Detection of SNPs in HER2, MUC1, ESR1, and BRCA1 genes associated with canine mammary cancer**. **Carvajal-Agudelo, J. D.**, Giraldo-Chalarca, L., Cortes-Mera, D. M., Ossa-López, P. A., Morales-Álvarez, E. D., & Rivera-Páez, F. A. *Veterinarska Stanica, 52(5).* doi:10.46419/vs.52.5.1

2021 **Two new feather mites of the genus *Trochilodectes* from hummingbirds in Colombia**. Hernandes, F. A., Castaño-Villa, G. J., **Carvajal-Agudelo, J. D.** & Rivera-Páez, F. A. *Systematic and Applied Acarology, 1495–1507.* doi:10.11158/saa.26.8.7

2021 **Periurban control barriers: the role of wild mammals in the circulation of hemoparasites in the Manizales (Caldas) ecopark network**. Rivera Páez, F., Ossa López, P., **Carvajal Agudelo, J.**, Trujillo Betancur, M., Mejía Fontecha, I., Banguero, L., Grueso, R., Serna, C., Tobón Escobar, W., Gallego Londoño, B., Reyes, S., Ocampo Grajales, J., Grajales Hahn, S., Escobar Aguirre, Y., Caicedo Guerrero, D., & Ramírez Chaves, H. *Research seedbed project, Universidad de Caldas.* repositorio.ucaldas.edu.co

2020 **New records and confirmation of three primate species in southwestern Colombia**. Ramírez-Chaves, H. E., **Carvajal-Agudelo, J. D.**, Hoyos R., M., Bustamante-Manrique, S., Castaño-Rivera, A., Rivillas-Carmona, M. A., Ossa-López, P. A., & Rivera-Páez, F. A. *Check List, 16(4): 831–839.* doi:10.15560/16.4.811

RESEARCH PROJECTS & GRANTS

Understanding the impacts of whole-body vibration exposure on bone health
Research Assistant. 2024–2025 · Research Nova Scotia. RT-qPCR · Tissue staining · Zebrafish.

Simulated microgravity & vibration effects in bone tissue in vivo
Research Assistant. 2023–2024 · Canadian Space Agency. Grant ~CAD 50K · Co-applicant. qPCR · Cell culture · Zebrafish larvae.

Microgravity-induced bone loss via simulated microgravity experiments
Research Assistant. 2022–2023 · Canadian Space Agency. Gene expression · Zebrafish scales.

Laboratory COVID-19
Molecular Analyst. 2021 · Ministerio de Ciencias Colombia & Universidad de Caldas. Molecular diagnostics · SARS-CoV-2 detection.

Implementation of an innovation system to address problems associated with high-risk biological agents to human health through the strengthening of laboratories in the Caldas department
Molecular Analyst. 2020 · Ministerio de Ciencias Colombia & Universidad de Caldas. Molecular diagnostics · Laboratory strengthening.

The role of wild birds and small mammals in the circulation of ticks and rickettsiae in the department of Arauca, Colombia
Master's student. 2019–2021 · Ministerio de Ciencias Colombia & Universidad de Caldas. 16S Metagenomics · Field sampling.

Peri-urban control barriers: hemoparasites in wild mammals of Manizales eco-parks
Master's student & co-researcher. 2019–2020 · Universidad de Caldas. Grant ~CAD 1K · Co-applicant. PCR · DNA sequencing · Field sampling.

***T. cruzi* & *T. rangeli* in wild mammals from Arauca, Colombia**
Undergraduate thesis co-supervisor. 2019–2021 · Universidad de Caldas. Molecular + morphological characterization.

HER2, MUC1 & ER-α gene expression in canine breast cancer early diagnosis
Undergraduate student. 2017–2020 · Universidad de Caldas. Grant ~CAD 10K · Co-applicant. SNPs · RT-qPCR · Epidemiological analysis.

AWARDS & RECOGNITION

NSERC Canada Graduate Scholarship, Doctoral (CGRS D)
Natural Sciences and Engineering Research Council of Canada. 2026. Federal doctoral scholarship supporting my PhD in Veterinary Medical Sciences at the University of Calgary.

Meritorious Thesis
Universidad de Caldas · Bacterial microbiome of *Ornithodoros cf. hasei* (Acari: Argasidae) and wild mammals in Arauca, Colombian Orinoquia. 2022. Distinction awarded to my Master's thesis in Biological Sciences on the bacterial microbiome of ticks and wild mammals in the Colombian Orinoquia.

ELAP Scholarship, Emerging Leader in the Americas Program
Global Affairs Canada. 2021. Scholarship that funded my research internship at Mount Allison University, Canada, working on *Borrelia* genotypes in Lloyd's Tick Lab.

Recognition for COVID-19 pandemic management
Consejo Profesional de Biología, Colombia. 2021. Recognition for my work as a Molecular Analyst detecting SARS-CoV-2 during the COVID-19 pandemic.

Master Scholarship
Ministerio de Ciencias Colombia & Universidad de Caldas. 2019–2021. Scholarship supporting my Master's studies in Biological Sciences at Universidad de Caldas (2019–2021).

Winner, Logo Design Contest
Biology Program, Universidad de Caldas. 2019. Won the contest to design the official logo of the Biology Program at Universidad de Caldas.

Leader of Undergraduate Research Group in Molecular Genetics (SeGeM)
Universidad de Caldas. 2017–2018. Led the undergraduate research group in molecular genetics, coordinating its members and research activities.

Fellow, Dreams of Peace Program
Bancolombia Foundation. 2015–2018. Scholarship that covered my full undergraduate Biology tuition from second year to graduation, plus a small stipend each semester.

Honour Academic Tuition
Universidad de Caldas. 2014 & 2016. Tuition waiver awarded for outstanding academic performance during my undergraduate studies. I received it twice: in 2014, for my second year, and in 2016, for my fourth year.

Award to Perseverance
Instituto Universitario de Caldas. 2012. Awarded by the Instituto Universitario de Caldas for the best average grades from 6th to 11th grade in high school.

SCIENTIFIC EVENTS & PRESENTATIONS

EVENT & PRESENTATION	ROLE	YEAR
12th Atlantic Regional Developmental Biology Symposium, Halifax, Canada, *Short-Term Gene Expression Responses in Danio rerio Larvae following Simulated Microgravity Exposure* **Carvajal-Agudelo J.D.** & Franz-Odendaal T.	**Oral Presenter**	2025

EVENT & PRESENTATION	ROLE	YEAR
Biology and Applied Aquatic Sciences (BAAS), Wolfville, NS, Canada, *Insights into 2D and 3D Simulated Microgravity: Impacts on Zebrafish Skeleton Ossification and Gene Expression* **Carvajal-Agudelo J.D.** & Franz-Odendaal T.	**Poster Presenter**	2025
1st ISCB LGBTQ+ Symposium, online (International Society for Computational Biology)	**Organizer**	2024
20th ISCB Student Council Symposium (SCS 2024), Montréal, Canada (published in *Bioinformatics Advances*)	**Organizer**	2024
Biology and Applied Aquatic Sciences (BAAS), Wolfville, NS, Canada, *Effects of mechanostimulation on adult bone homeostasis in Zebrafish* Joseph F., **Carvajal-Agudelo J.D.** & Franz-Odendaal T.	**Co-author**	2024
Biology and Applied Aquatic Sciences (BAAS), Wolfville, NS, Canada, *Gene expression changes in larval zebrafish (Danio rerio) after microgravity and vibration exposure* LeDrew E., **Carvajal-Agudelo J.D.** & Franz-Odendaal T.	**Co-author**	2024
Canadian Space Health Research Symposium, London, Ontario, Canada, *Short-Term Gene Expression Responses in Danio rerio Larvae following Simulated Microgravity Exposure* **Carvajal-Agudelo J.D.** & Franz-Odendaal T.	**Poster Presenter**	2024
11th Atlantic Regional Developmental Biology Symposium, Halifax, Canada, *Exploring the Effects of Simulated Microgravity in Zebrafish: Differential Gene Expression and Ossification in 2D vs 3D Rotation* **Carvajal-Agudelo J.D.**, Eaton J. & Franz-Odendaal T.	**Oral Presenter**	2024
American Association for Anatomy Annual Meeting, Washington DC, USA, *Spatial and Temporal Patterning of the Tracheal Cartilage* Franz-Odendaal T., **Carvajal-Agudelo J.D.**, Apienti T., Tench L. & Muntzar R.	**Co-author**	2023
QAtCanSTEM, MSVU Halifax, Canada, *Effects of simulated microgravity and vibration on osteoblast and osteoclast activity in cultured zebrafish scales* **Carvajal-Agudelo J.D.**, McNeil A. & Franz-Odendaal T.	**Poster Presenter**	2022
9th Atlantic Regional Developmental Biology Symposium, MSVU Halifax, Canada, *Does exposure to simulated microgravity and vibration affect bone cells in a zebrafish scale system?* **Carvajal-Agudelo J.D.**, McNeil A. & Franz-Odendaal T.	**Oral Presenter**	2022
4th RSG-Colombia Bioinformatics Symposium (ISCB Student Council RSG Colombia), Universidad de los Andes, Caribbean Campus, Cartagena, Colombia	**Organizer**	2022
3rd RSG-Colombia Bioinformatics Symposium (ISCB Student Council RSG Colombia), virtual, round table "Professional prospects in bioinformatics: between academia and industry"	**Organizer**	2021
XVII Colombian Congress of Parasitology and Tropical Medicine, Cali, Colombia, *Wild mammals involved in the circulation of Rickettsia spp. in the department of Arauca (Colombian Orinoquia)* Velásquez-Guarín D., Mejía-Fontecha I.Y., **Carvajal-Agudelo J.D.**, Rivera-Páez F.A., Pérez Cárdenas J.E. & Ramírez-Chaves H.E.	**Co-author**	2019
XVII Colombian Congress of Parasitology and Tropical Medicine, Cali, Colombia, *Molecular characterization and morphological detection of Trypanosoma spp. in wild mammals from Arauca, Colombia: an approach to potential reservoirs* Trujillo-Betancur M.P., **Carvajal-Agudelo J.D.**, Ramírez-Chaves H.E. & Rivera-Páez F.A.	**Co-author**	2019
XVII Colombian Congress of Parasitology and Tropical Medicine, Cali, Colombia, *Comparative evaluation of DNA conservation and extraction methods in blood of wild mammals used in the study of hemoparasites* **Carvajal-Agudelo J.D.**, Trujillo-Betancur M.P., Velásquez-Guarín D., Pérez Cárdenas J.E., Ramírez-Chaves H.E. & Rivera-Páez F.A.	**Poster Presenter**	2019
VIII Colombian Symposium of Virology / IV Latin American Congress of Virology, Bogotá, Colombia, *HIV mortality and TB coinfection in Caldas, Colombia (2010–2018)* **Carvajal-Agudelo J.D.**, Rivera-Páez F.A. & Pérez Cárdenas J.E.	**Poster Presenter**	2019
Bat Week (Semana del Murciélago), Museum Center, Universidad de Caldas, Manizales, Colombia, exhibition and talks on bats	**Organizer**	2018
XV Colombian & IX International Human Genetics Congress, Barranquilla, Colombia, *Epidemiological meta-analysis of clinical variables related to canine breast cancer* **Carvajal-Agudelo J.D.** & Cortes Mera D.M. (other authors)	**Co-author**	2018
V Regional Biology Symposium, UTP, Santa Rosa de Cabal, Colombia, *Evaluation of the genotoxic effect induced by phenthoate in Poecilia reticulata (Cyprinodontiformes: Poeciliidae)* **Carvajal-Agudelo J.D.** & Marulanda S. (and others)	**Oral Presenter**	2018

EVENT & PRESENTATION	ROLE	YEAR
V Regional Biology Symposium, UTP, Santa Rosa de Cabal, Colombia, Hot spring microbial ecology **Carvajal-Agudelo J.D.** & Marulanda S. (and others)	**Poster Presenter**	2018

POSTGRADUATE COURSES & TRAINING

Bioinformatics, data science & programming

COURSE	INSTITUTION	YEAR
Bioinformatics Specialization (3 courses): Finding Hidden Messages in DNA; Genome Sequencing; Comparing Genes, Proteins, and Genomes	University of California, San Diego – Coursera	2018–2019
Bioinformatic Methods I & II	University of Toronto – Coursera	2020
Bioinformatics for Biologists: An Introduction to Linux, Bash Scripting, and R	Wellcome Connecting Science – FutureLearn	2023
Introduction to Systems Biology	Icahn School of Medicine at Mount Sinai – Coursera	2016
What is Data Science?	IBM – Coursera	2020

Genomics & molecular diagnostics

COURSE	INSTITUTION	YEAR
Genomic Data Science Specialization: Introduction to Genomic Technologies; Bioconductor for Genomic Data Science	Johns Hopkins University – Coursera	2019–2020
Bacterial Genomes Program (5 courses): DNA to Protein Function; Accessing and Analysing Microbial Genome Data; Disease Outbreaks and Antimicrobial Resistance; Comparative Genomics with ACT; Antimicrobial Resistance in Bacterial Pathogens	Wellcome Genome Campus – FutureLearn	2018–2020
Metagenomics applied to surveillance of pathogens and antimicrobial resistance	Technical University of Denmark – Coursera	2019
Whole Genome Sequencing of Bacterial Genomes: Tools and Applications	Technical University of Denmark – Coursera	2020
Genomic Technologies in Clinical Diagnostics: Molecular Techniques	St George's, University of London – FutureLearn	2018
Analysis of Exomes in Cancer (pre-congress course, Cali, Colombia)	Colombian Congress on Computational Biology	2017
Evolutionary Genomic Training at the Caribbean Coast	Universidad de los Andes Sede Caribe, Cartagena	2019
Genetic Engineering and its Application to Microbial Biotechnology (22 h)	Universidad Santiago de Cali	2018
Basic Molecular Biology Module 4: PCR and Real-Time PCR	CDC	2020
Workshop on Molecular Detection of Pathogens (5 h)	IEEE EMBS Student Chapter ESPE	2020
Histology and Molecular Biology of Ticks (40 h)	GEBIOME, Universidad de Caldas – Brazilian Central of Studies on Ticks Morphology (UNESP)	2016

Parasitology, epidemiology & global health

COURSE	INSTITUTION	YEAR
Mammal Parasitology: Field Methods (Centro de Investigación La Palmita, Casanare, Colombia)	H. W. Manter Laboratory of Parasitology, University of Nebraska-Lincoln – Universidad de Antioquia	2019
Tropical Parasitology: Protozoans, Worms, Vectors and Human Diseases	Duke University & Kilimanjaro Christian Medical University College – Coursera	2019
Epidemiology: The Basic Science of Public Health	University of North Carolina at Chapel Hill – Coursera	2019
COVID-19: Tackling the Novel Coronavirus	London School of Hygiene & Tropical Medicine – FutureLearn	2020
Global Health at the Human-Animal-Ecosystem Interface	University of Geneva – Coursera	2025

Ecology, GIS & fieldwork

COURSE	INSTITUTION	YEAR
Geographic Information Systems (GIS) Specialization (5 courses)	University of California, Davis – Coursera	2018–2019
Diploma in ICT Competencies: Data Collection and Georeferencing (120 h)	Corporación Unificada Nacional (CUN) & Colnodo – Google.org JuvenTIC project	2018
Biological Diversity: Theories, Measures and Data Sampling Techniques	National Research Tomsk State University – Coursera	2020
Global Environmental Management	Technical University of Denmark – Coursera	2025
Wilderness First Aid: Introduction	University of Colorado System – Coursera	2025

University teaching & scientific writing

COURSE	INSTITUTION	YEAR
Diploma in University Teaching and Didactics	Politécnico Superior de Colombia, Medellín	2020
Scientific Reading and Writing (16 h)	Universidad de Caldas, Vice-Rectorate for Research and Postgraduate Studies	2016

Animal care & research ethics

CERTIFICATE	INSTITUTION	YEAR
Introduction to the Care and Use of Fish (NIAUT requirement)	Dalhousie University – CCAC / NIAUT	2022
CCAC Online Ethics – Fish Users	University of British Columbia	2025
Research Ethics – Tri-Council Policy Statement (TCPS 2: CORE 2022)	Panel on Research Ethics, Government of Canada	2022
From Tank to Paper: Advance Reproducibility through Better Reporting (webinar)	Zebrafish Husbandry Association	2025
Ethics in University Research (36 h)	Universidad del País Vasco – MiriadaX	2016

VOLUNTEERING & MEMBERSHIPS

Volunteering

Mentor, Graduate Leaders Circle (GLC) for NSERC CGRS D applicants
University of Calgary. 2026 – Present. Mentor students preparing their NSERC doctoral scholarship applications, sharing my experience as an NSERC CGRS D winner.

Volunteer, Kids Saving the Rainforest (KSTR)
2026 – Present. Volunteered for one month in May at a wildlife sanctuary and clinic in Costa Rica, supporting the care and rehabilitation of rescued wild animals. I now continue as a grant fundraising volunteer, researching and applying for grants.

Volunteer Maker, Clubs That Care (Makers Making Change)
Neil Squire Society, built assistive technology devices for people with disabilities. 2025. Helped build low-cost assistive technology devices for people with disabilities through the Makers Making Change program of the Neil Squire Society.

Secretary, RSG Colombia, ISCB Student Council
rsgcolombia.github.io. 2021–2024. Served as Secretary of the Regional Student Group of Colombia of the ISCB Student Council, supporting its activities and co-organizing its bioinformatics symposia.

Outreach Volunteer, ISCB Student Council (ISCB SC)
International Society for Computational Biology (ISCB) Student Council. 2025. Supported outreach activities of the Student Council, helping to share its programs and events with the computational biology student community.

Volunteer Designer, ISCB
International Society for Computational Biology (ISCB). 2026. Designed graphic materials for ISCB activities and communications.

Social Media Manager & Volunteer
SAVE Mexico, Turtle Conservation, Akumal, Mexico. 2023. Managed social media and outreach for a sea turtle conservation organization, and supported conservation activities in Akumal, Mexico.

General Communication Manager
Orquídeas para la Paz, The Arias Lab, Antioquia, Colombia · theariaslab.github.io. 2022–2024. Led general communications for Orquídeas para la Paz, an orchid conservation initiative of The Arias Lab, sharing its work with the public.

Curriculum Representative, Master's in Biological Sciences
Universidad de Caldas. 2020–2021. Represented students in the curriculum matters of the Master's program, acting as the link between students and faculty.

Memberships

Member, Innovation Sandbox
Hunter Hub for Entrepreneurial Thinking · University of Calgary. 2026 – Present. Training and mentorship program in entrepreneurship and innovation.

Junior Researcher (Investigador Junior), SNCTeI
Ministerio de Ciencia, Tecnología e Innovación (Minciencias), Colombia. 2025 – Present. Officially recognized within the National System of Science, Technology and Innovation (SNCTeI) of Colombia. Scientific evaluator for Convocatoria 975 de 2026, reviewing master's-level research proposals.

Member, ISCB
International Society for Computational Biology (ISCB). 2019 – Present. International professional society for computational biology and bioinformatics.

Member, RSG Colombia
Regional Student Group (RSG) Colombia, part of the ISCB Student Council. 2019 – Present. Colombian chapter of the ISCB Student Council, a community of students and young researchers in bioinformatics and computational biology.

Member, GEBIOME Research Group
Genetics, Biodiversity and Ecosystem Management · Universidad de Caldas. 2017–Present. Research group focused on genetics, biodiversity and ecosystem management; collaborated on research projects in these areas.

Member, Genomics and Conservation Research Group
The Colombian Orchid Society. 2020–2021. Research group applying genomic tools to the conservation of Colombian orchids.

Member & Leader, SeGeM (Molecular Genetics Research Group)
Universidad de Caldas. 2017–2023. Molecular genetics research group where I began my research training, worked on molecular and genetic studies, and served as group leader (2017–2018).

Member, SIMAS (Mammalogy Research Group)
Semillero de Investigación en Mastozoología · Universidad de Caldas. 2017–2019. Undergraduate research group in mammalogy at the Universidad de Caldas.

Member, Natural History Laboratory: Integrative Zoological Biodiversity Discovery (IZBD)
Museo de Historia Natural, Centro de Museos · Universidad de Caldas. 2019–2021. Research laboratory of the Natural History Museum of the Universidad de Caldas, focused on zoological biodiversity; joined during my Master's studies.

Member, American Genetic Association (AGA)
American Genetic Association. 2019–2020. US scientific organization dedicated to genetics and genomics, founded in 1903 and publisher of the *Journal of Heredity*.

Member, Colombian Association of Parasitology and Tropical Medicine (ACPMT)
Asociación Colombiana de Parasitología y Medicina Tropical. 2019–2020. Colombian scientific association founded in 1965 that brings together parasitology researchers and organizes the biennial Colombian Congress of Parasitology and Tropical Medicine.

Member, ABEUC
Asociación de Biólogos Egresados de la Universidad de Caldas (Association of Biology Graduates of the Universidad de Caldas). 2019 – Present. Professional association of biologists who graduated from the Universidad de Caldas, connecting alumni across the profession.